

Text with punctuation padded

```
wine review : US : California :  
Pinot Noir : Oak and earth  
intermingle around robust aromas  
of wet forest floor in this  
vineyard - designated Pinot that  
bails from a high - elevation  
site . Small in production , it  
offers intense , full - bodied  
raspberry and blackberry steeped  
in smoky spice and smooth  
texture .
```

Tokenized text

```
[ 7 10 2 20 2 28 2 43 62 2 55  
5 243 4145 453 634 26 9 497  
499 667 17 12 142 14 2214 43  
25 2484 32 8 223 14 2213 948 4  
594 17 987 3 15 75 237 3 64 14  
82 97 5 74 2633 17 198 49 5  
125 77 4 0 0 0 0 0 0 0 0 0  
0 0 0 0 0 0 0 0 0 0 0]
```

Input

Training observation

```
<tf.Tensor: shape=(80,), dtype=int64, numpy=  
array([ 7, 10, 2, 42, 2, 65, 2, 45, 14, 43,  
24, 27, 2, 46, 65, 3, 12, 13, 8, 523, 3,  
2464, 7, 25, 372, 116, 57, 22, 3, 107, 773,  
5, 553, 49, 4, 6, 120, 13, 79, 3, 11, 34, 5,  
8, 58, 249, 3, 33, 15, 18, 21, 6, 68, 3,  
263, 22, 25, 13, 91, 1192, 4, 0, 0, 0, 0,  
0, 2, 8, 0, 0, 0, 0, 0, 8, 8, 0, 0, 0, 0])>
```

Output: shift by one token

```
<tf.Tensor: shape=(80,), dtype=int64, numpy=  
array([ 10, 2, 42, 2, 65, 2, 45, 14, 43, 24,  
27, 2, 46, 65, 3, 12, 13, 8, 523, 3, 2464,  
7, 25, 372, 116, 57, 22, 3, 107, 773, 5,  
553, 49, 4, 6, 120, 13, 79, 3, 11, 34, 5, 8,  
58, 249, 3, 33, 15, 18, 21, 6, 68, 3, 263,  
22, 25, 13, 91, 1192, 4, 0, 0, 0, 0, 8, 8,  
0, 2, 8, 0, 0, 0, 0, 0, 8, 8, 0, 0, 0, 0])>
```

Figure: Data processing for the Transformer

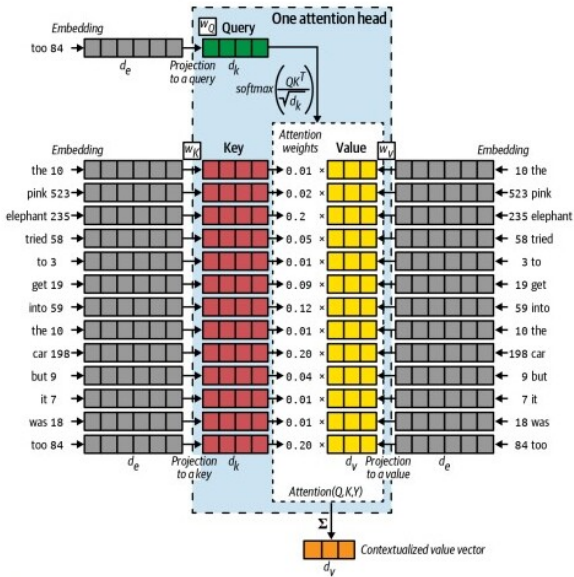


Figure: The mechanics of an attention head

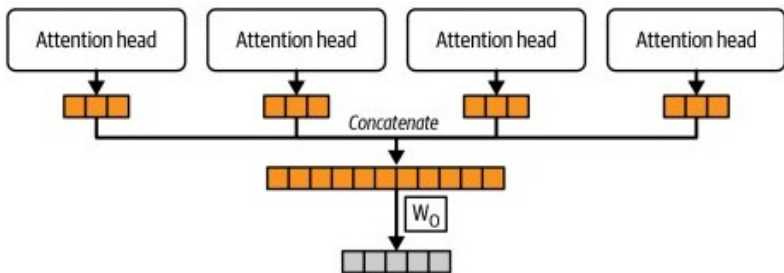


Figure: A multihead attention layer with four heads

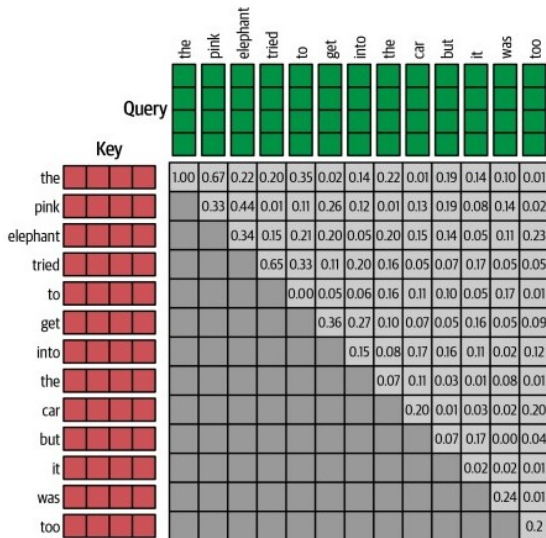


Figure: Matrix calculation of the attention scores for a batch of input queries, using a causal attention mask to hide keys that are not available to the query (because they come later in the sentence)

```
array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [0, 1, 1, 1, 1, 1, 1, 1, 1, 1],
       [0, 0, 1, 1, 1, 1, 1, 1, 1, 1],
       [0, 0, 0, 1, 1, 1, 1, 1, 1, 1],
       [0, 0, 0, 0, 1, 1, 1, 1, 1, 1],
       [0, 0, 0, 0, 0, 1, 1, 1, 1, 1],
       [0, 0, 0, 0, 0, 0, 1, 1, 1, 1],
       [0, 0, 0, 0, 0, 0, 0, 1, 1, 1],
       [0, 0, 0, 0, 0, 0, 0, 0, 1, 1],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 1]], dtype=int32)
```

Figure: The causal mask as a numpy array—1 means unmasked and 0 means masked

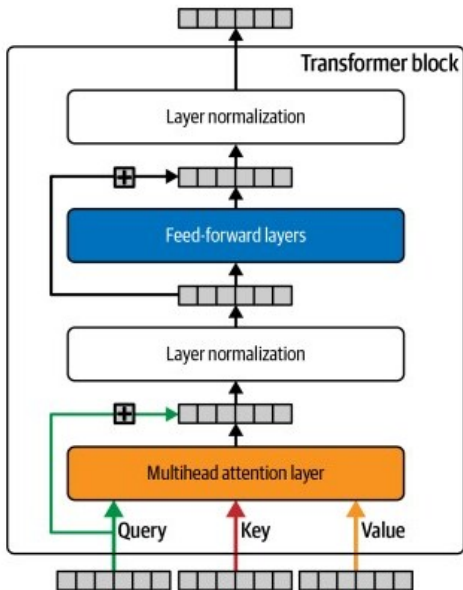


Figure: A Transformer block

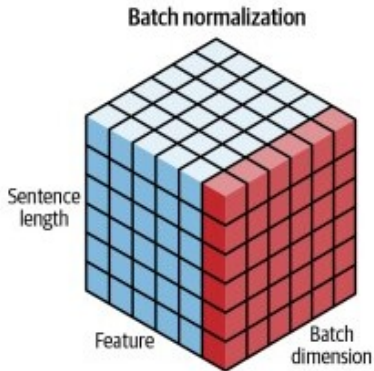
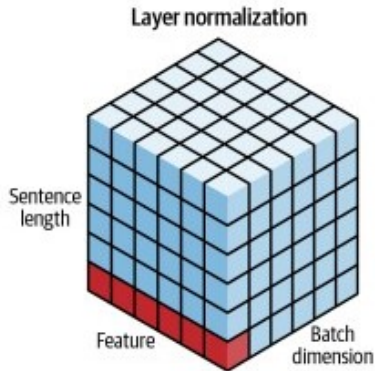


Figure: Layer normalization versus batch normalization—the normalization statistics are calculated across the blue cells

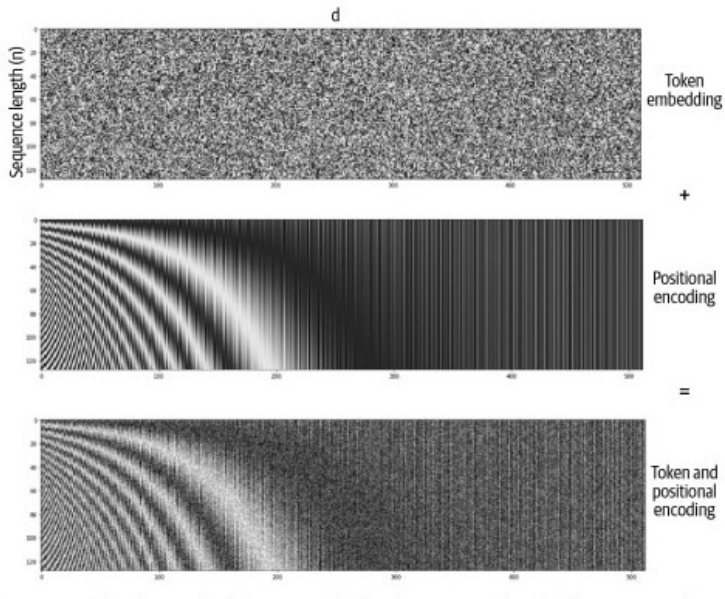


Figure: The token embeddings are added to the positional embeddings to give the token position encoding

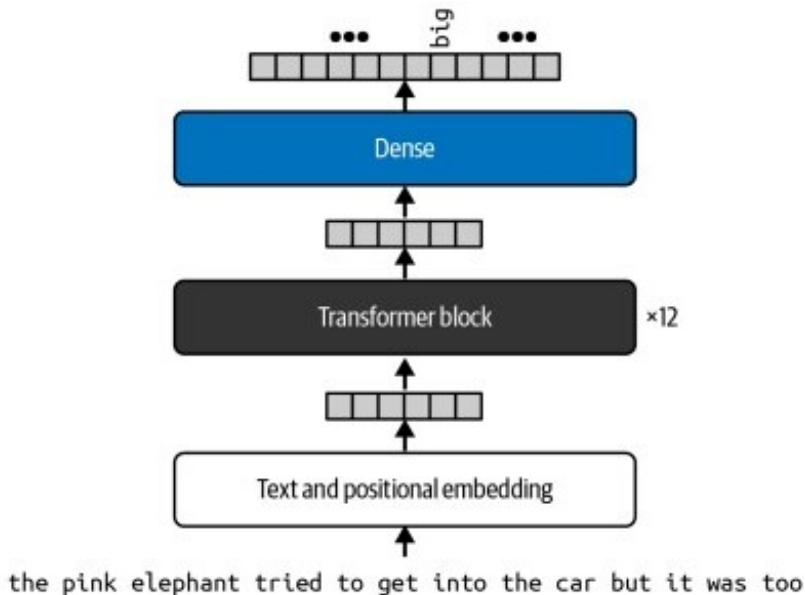


Figure: The simplified GPT model architecture

temperature = 1.0

Generated text:

wine review : us : washington : chenin blanc : a light , medium - bodied wine , this light - bodied expression is not a lot of enjoyment . it ' s simple with butter and vanilla flavors that frame mixed with expressive fruit . it ' s juicy and tangy with a lemon lingers on the finish .

temperature = 0.5

Generated text:

wine review : italy : piedmont : nebbiolo : this opens with aromas of french oak , menthol and a whiff of toast . the straightforward palate offers red cherry , black raspberry jam and a hint of star anise alongside firm but rather fleeting tannins , drink through 2016 .

Figure: Generated outputs at temperature = 1.0 and temperature = 0.5.

wine review : germany :

pfalz:	51.53%
mosel:	41.21%
rheingau:	4.27%
rheinhessen:	2.16%
franken:	0.44%

wine review : germany : rheingau : riesling : this is a ripe , full - bodied

riesling:	46.56%
,:	27.78%
wine:	16.88%
and:	4.58%
yet:	1.33%

wine review : germany : rheingau : riesling : this is a ripe , full - bodied riesling
with a touch of residual sugar , it ' s a slightly

sweet:	94.23%
oily:	1.25%
viscous:	1.09%
bitter:	0.88%
honeyed:	0.66%

Figure: Distribution of word probabilities following various sequences

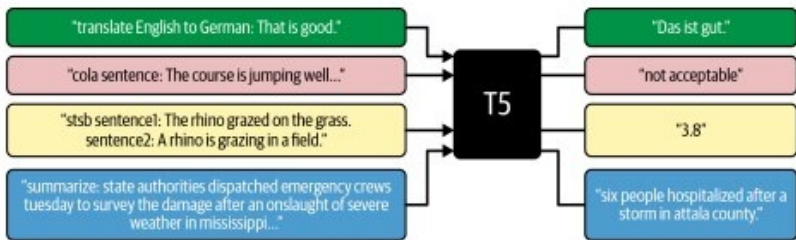


Figure: Examples of how T5 reframes a range of tasks into a text-to-text framework, including translation, linguistic acceptability, sentence similarity, and document summarization

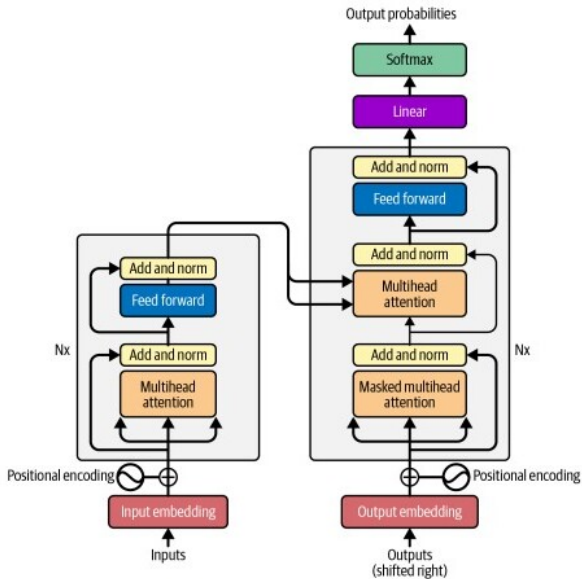


Figure: An encoder-decoder Transformer model: each gray box is a Transformer block

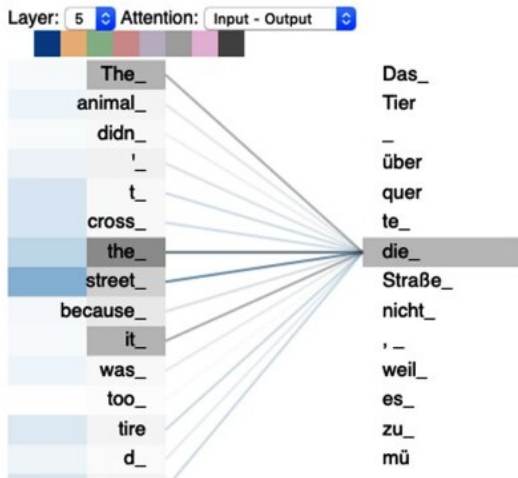


Figure: An example of how one attention head attends to the word “the” and another attends to the word “street” in order to correctly translate the word “the” to the German word “die” as the feminine definite article of “Straße”

Write me a short story about a unicorn who defeats an evil dragon.

Luna was out for a leisurely stroll through the forest when she heard a loud, ferocious roar. She immediately knew it was a dragon and ran as fast as she could in the opposite direction.

As she ran, she could feel the dragon's hot breath on her back and knew she wouldn't be able to outrun it for long. She had to think of a plan, and fast.

Just as the dragon was about to snatch her up in its teeth, Luna turned around and stabbed it in the eye with her horn. The dragon screeched in pain and flew into a rage, but Luna was ready for it.

She dodged its fire breath and claws until she finally had an opportunity to strike. She impaled the dragon with her horn and it fell to the ground with a thud, dead.

Luna breathed a sigh of relief and trotted away, knowing that she had saved the forest from the evil dragon.

Figure: An example of how GPT-3 can extend a given system prompt

What is a Transformer model and how is it used in generative deep learning? Can you give a short ~100 word summary?



A Transformer model is a type of deep learning model used in natural language processing tasks such as language translation and text generation. It utilizes an attention mechanism to enable the model to focus on different parts of the input sequence during processing, allowing it to capture long-range dependencies and better model context.



In generative deep learning, a Transformer model can be used as a language model to generate new text that is similar to the input data it was trained on. This can be achieved by using the model's output at each time step as input to the next time step, and sampling from the output probability distribution to generate the next word in the sequence. With sufficient training data and appropriate tuning of hyperparameters, Transformer models can produce high-quality, coherent text that is indistinguishable from text written by humans.



What's attention? Reply in the form of a limerick.



Attention's a powerful tool
For deep learning models to rule
It lets them decide
Which parts to abide
And focus on features that are cool



If a Transformer were an animal, what would it be and why?



If a Transformer were an animal, it would likely be a chameleon.



Like a Transformer, a chameleon has the ability to adapt and change its appearance to blend in with its environment, allowing it to be more effective in different situations. Similarly, a Transformer model can adapt its attention to focus on different parts of input sequences based on the specific task it is performing, allowing it to excel at a wide range of natural language processing tasks.

In addition, both chameleons and Transformers have a reputation for being able to "blend in" seamlessly with their surroundings, making them difficult to detect. In the case of Transformers, this is often because the generated text is so coherent and similar to human-written text that it can be difficult to distinguish between the two.

Figure: An example of ChatGPT answering questions about Transformers

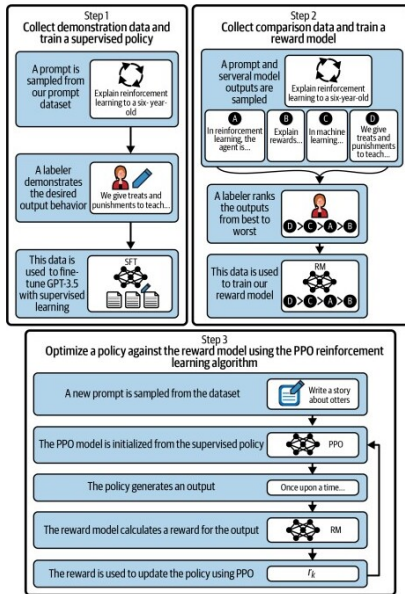


Figure: The reinforcement learning from human feedback fine-tuning process used in ChatGPT